

Assignment 3 - final

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Assignment 3

- Exercise 1 - Properties of Gold codes, pt. 10
- Exercise 2 - Impact of cross-correlation on CDMA performance, pt. 12
- Exercise 3 - Hartogs-Hughes algorithm for OFDM bit allocation, pt. 8

Gold sequences: construction

- An m-sequence c_1 generated by a Linear Feedback Shift Registers (LFSR) with m cells and primitive polynomial $p_1(D)$.
- Another m-sequence c_2 generated by a Linear Feedback Shift Registers (LFSR) with m cells and primitive polynomial $p_2(D)$ paired to $p_1(D)$.
- The set of Gold codes is made by the $(2^m + 1)$ sequences obtained as:

$$\begin{cases} c_1 \\ c_2 \\ c_1 + T^i(c_2) \quad 0 \leq i < 2^m - 1 \end{cases}$$

where T^i is the cyclic shift by i positions and $+$ is the binary sum (ex-or) in $GF(2)$.

Gold sequences: properties

Basic properties:

- Sequence length (period):

$$N = 2^m - 1$$

- Number of Gold sequences:

$$2^m + 1$$

- cross-correlation property: three possible values

$$\begin{cases} -t \\ -1 \\ t - 2 \end{cases}$$

$$t = 2^{\frac{m+1}{2}} + 1 \text{ for } m \text{ odd, } t = 2^{\frac{m+2}{2}} + 1 \text{ for } m \text{ even}$$

Gold sequences: Details

Let α be a primitive element of $\text{GF}(2^m)$ and $p_1(D)$ the minimal polynomial of α .

The paired polynomial $p_2(D)$ is the minimal polynomial of α^t where:

- $t = 2^{\frac{m+1}{2}} + 1$ for m odd
- $t = 2^{\frac{m+2}{2}} + 1$ for m even

Equivalently, the sequence c_2 is obtained by sampling the (periodic) sequence c_1 every t bits:

$$u_2(n) = u_1(nt) \quad 0 \leq n < 2^m - 1$$

Gold sequences: Polynomials

m	N	p_1	p_2
7	127	[7 4 0]	[7 1 0]
10	1023	[10 3 0]	[10 8 3 2 0]

Gold sequences: Matlab

Matlab Gold function

```
m=7;  
L=2^m-1;  
goldseq = comm.GoldSequence('FirstPolynomial','x^7+x^4+1',...  
'SecondPolynomial','x^7+x+1',...  
'FirstInitialConditions',[0 0 0 0 0 0 1],...  
'SecondInitialConditions',[0 0 0 0 0 0 1],...  
'Index',1,'SamplesPerFrame',L);  
x1 = goldseq();  
x1b=1-2*x1;
```

Index

Index = -2 $\Rightarrow c_1$

Index = -1 $\Rightarrow c_2$

Index = i ($0 \leq i < 2^m - 1$) $\Rightarrow c_1 + T^i(c_2)$

Definitions

Given a set $\{c_i\}_{i=1}^K$ of K bipolar sequences of period N we define:
The cyclic autocorrelation function

$$r_{c'_i}(\tau) = \sum_{n=1}^N c'_i(n) c'_i(n + \tau) \quad 0 \leq \tau < N$$

The cyclic crosscorrelation function

$$r_{c'_i c'_j}(\tau) = \sum_{n=1}^N c'_i(n) c'_j(n + \tau) \quad 0 \leq \tau < N$$

Definitions

Given a set $\{\underline{c}'_i\}_{i=1}^K$ of K bipolar sequences of period N we define

- the maximal out-of-phase cyclic auto-correlation magnitude

$$r_A = \max \left\{ \left| r_{c'_i}(\tau) \right| \quad 1 \leq \tau \leq N-1 \quad 1 \leq i \leq K \right\}$$

- the maximal cyclic cross-correlation magnitude

$$r_C = \max \left\{ \left| r_{c'_i c'_j}(\tau) \right| \quad 0 \leq \tau \leq N-1 \quad 1 \leq i, j \leq K \quad i \neq j \right\}$$

- the maximal cyclic correlation magnitude

$$r_M = \max \{ r_C, r_A \}$$

Welch bound

It holds for complex sequences of symbols with unitary magnitude:

$$r_M \geq N^{2s} \sqrt{\frac{1}{KN-1} \left[\frac{KN}{\binom{N+s-1}{s}} - 1 \right]} \quad \forall s \in \mathbb{Z}^+ = \{1, 2, 3, \dots\}$$

(you can stop at $s=10$, if the root argument is negative set it to 0)

It is often simplified as:

$$r_M \geq N \sqrt{\frac{K-1}{KN-1}}$$

that, for large K , can be approximated to:

$$\sqrt{N}$$

(Since we are working with binary bipolar sequences, use ceil.)

Sidelnikov bound

It holds for binary bipolar sequences:

$$r_M \geq \left\lfloor \sqrt{\left[(2s+1)(N-s) + \frac{s(s+1)}{2} - \frac{2^s N^{2s+1}}{K(2s)! \binom{N}{s}} \right]} \right\rfloor$$
$$\forall s \in \mathbb{N} = \{0, 1, 2, 3, \dots\} \quad 0 \leq s < \frac{2N}{5}$$

(you can stop at $s=10$, if the root argument is negative set it to 0)

Since the Sidelnikov bound applies to binary sequences, it is tighter than the Welch bound.

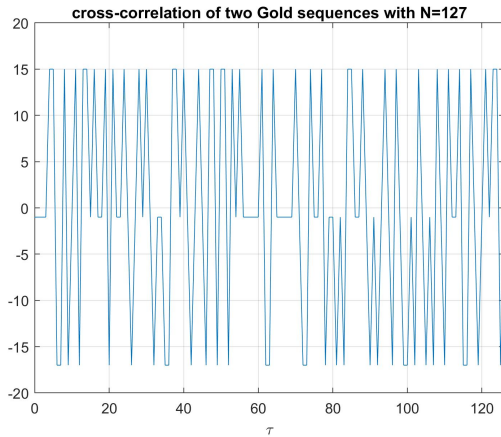
Exercise 1 - part 1 - pt. 2

Fix $m = 7$

- 1 Generate a Gold sequences with Index $i \geq 0$
- 2 Compute and plot the cyclic auto-correlation
- 3 Repeat for $i < 0$
- 4 Comment the results

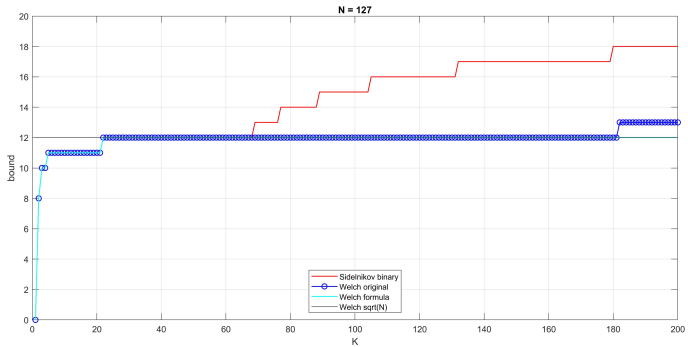
Exercise 1 - part 2 - pt. 1

- ⑤ Generate two random Gold sequences
- ⑥ Compute and plot their cyclic cross-correlation
- ⑦ Repeat when one of the two sequences has Index $i < 0$
- ⑧ Comment the results



Exercise 1 - part 3 - pt. 4

- 9 Plot the three versions of the Welch bound and the Sidelnikov bound for $N = 127$ and $1 \leq K \leq 200$
- 10 Consider the entire set of $K = 129$ Gold sequences and compute r_M (explain how you computed its value).
- 11 On the bound figure, add a point corresponding to r_M .
- 12 Comment the result



Exercise 1 - part 4 - pt. 3

- 13 Starting from the Gold set of $K = 129$ sequences, reduce the length of the sequences to $N = 126$. Compute r_M (explain how you computed its value).
- 14 Compute and plot the bounds for $N = 126$ and $K = 129$.
- 15 Add r_M to the plot.
- 16 Comment the result.

Exercise 2 - CDMA performance and cross-correlation

Important: This is an exercise on CDMA.

The first user U_1 generates a random bit v_1 , converts it into a bipolar symbol v'_1 and use $\underline{c}'_1$ ($N=8$ chips) as spreading sequence.

The channel adds noise and we receive a vector $\underline{r}$ of N real values.

We project $\underline{r}$ over $\underline{c}'_1$, we take a decision based on the sign and we compare with v_1 .

This is the first scenario (CDMA with a single user).

In the second scenario, there is also a second user U_2 that generates a random bit v_2 , converts it into a bipolar symbol v'_2 and use $\underline{c}'_2$ ($N=8$ chips) as spreading sequence.

Now the received vector $\underline{r}$ of N real values is obtained by summing the signals transmitted by U_1 and U_2 and the noise. Again, we project $\underline{r}$ over $\underline{c}'_1$, we take a decision based on the sign and we compare with v_1 . This is the second scenario (CDMA with two users)

Exercise 2 - part 1 - pt. 2

First scenario. Fix the length of the spreading sequences to $N = 8$ chips.

- 1 Generate a random bipolar spreading sequence $\underline{c}'_1$ of N chips.
- 2 Simulate the error rate performance of a single user U_1 transmitting its random information bits by using $\underline{c}'_1$ as spreading sequence for each information bit.
- 3 Since there is a single user transmitting with no interference, the system error rate is equivalent to a 2-PAM (with basis signal equal to $\underline{c}'_1$). Verify that the BER is equal to:

$$P(e) = \frac{1}{2} \operatorname{erfc} \sqrt{\frac{E_b}{N_0}}$$

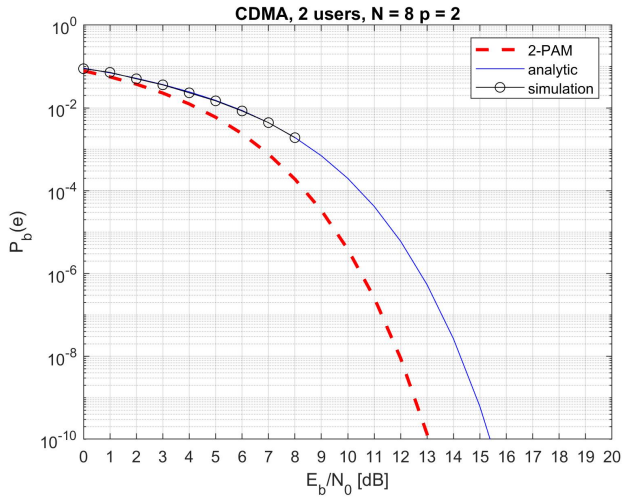
Exercise 2 - part 2 - pt. 4

Second scenario.

- 4 Generate a second random bipolar spreading sequences $\underline{c}'_2$.
- 5 Now U_1 generates its random information bits and uses $\underline{c}'_1$ as spreading sequence, while U_2 generates its random information bits and used $\underline{c}'_2$ as spreading sequences. At the receiver side, recover the information bits transmitted by U_1 only, by projecting again on $\underline{c}'_1$. Simulate the performance of this CDMA system with two users using $\underline{c}'_1$ and $\underline{c}'_2$ without cyclic shift.
- 6 Add the error rate curve to the previous figure.
- 7 Which is the parameter p dominating the performance of the two-user scenario?
- 8 In the report present some use cases where you present the two binary spreading sequences, the value of p and the figure. Comment the results.

Exercise 2 - part 3 - pt. 3

- 9 In the report, derive analytically the performance curve of this CDMA system with two users (and no cyclic shift).
- 10 Add the analytic curves to the previous figures and verify that they are matched to the simulated curves
- 11 Discuss the results: why we have this behavior?



Exercise 2 - part 4 - pt. 3

What happens if we consider a realistic scenario where the two spreading sequences are received with a random cyclic shift, too?

- 12 Clearly explain which is the dominating parameter of the two-user performance in this case.
- 13 Derive the analytic curve in this case, too, and verify the match between simulated and analytic curves.

Exercise 3 - Hartogs - Hughes algorithm

Consider an OFDM system with $\Delta = 1$, $N = 128$ and carrier frequencies $f_i = i\Delta$.

Consider the following impulse response:

$$h(t) = a_1\delta(t - D_1) + a_2\delta(t - D_2) + a_3\delta(t - D_3) + a_4\delta(t - D_4)$$

where the delays are

$$D_1 = 1 \quad D_2 = 1.01 \quad D_3 = 1.015 \quad D_4 = 1.02$$

and the amplitudes are complex values with absolute values

$$|a_1| = 1 \quad |a_2| = 0.5 \quad |a_3| = 0.9 \quad |a_4| = 0.3$$

and random phase uniformly distributed between 0 and 2π .

Given $h(t)$ compute $H(f)$ and focus on the values $H_i = H(f = f_i)$.

Compute the attenuation values $A_i = 1/|H_i|^2$.

To apply the Hartogs Hughes algorithm, suppose the normalized attenuations are equal: $\alpha_i = A_i$ and the total power is $P_{TX} = \text{median}(A_i) * N$.

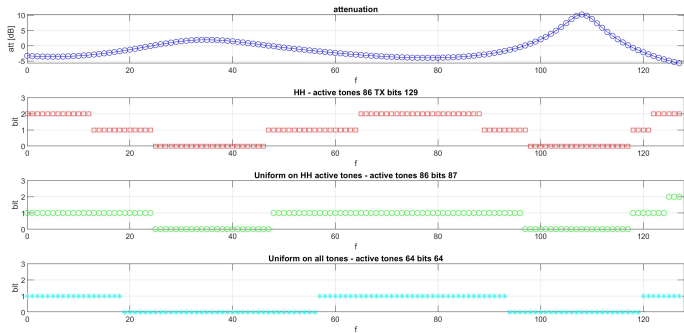
Exercise 3 - part 1 - pt. 5

- 1 Plot the attenuation profile in dB
- 2 Apply the Hartogs-Hughes algorithm, plot the number of bits transmitted on each tone, in the title write the number of active tones and the total number of transmitted bits.

Since the attenuation profile depends on the random phases, in the report present at least two figures obtained with two different channel realizations.

Exercise 3 - part 2 - pt. 3

- 3 Focus on the active tones identified with the HH algorithm, distribute P_{TX} uniformly over them, plot the number of bits transmitted on each tone, in the title write the number of active tones and the total number of transmitted bits.
- 4 Distribute P_{TX} uniformly over all the N tones, plot the number of bits transmitted on each tone, in the title write the number of active tones and the total number of transmitted bits.
- 5 Comment the results.



Deadline for +2 bonus: 22/12/2022 11 PM

Changes:

Exercise 2 improved description, and new figure with $N=8$